

Problem 3:

$$a) \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 3 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1/3 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/3 \end{bmatrix}$$

$$b) \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & 1 & -2 \\ 0 & 1 & 0 & 1 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$c) \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$d) \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ -5 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 5 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Problem 4:

$$\begin{aligned}
 (AB)^{-1} &= \begin{bmatrix} 1 & 6 \\ 0 & 3 \end{bmatrix}^{-1} \\
 &= \frac{1}{3} \begin{bmatrix} 3 & -6 \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & -2 \\ 0 & 1/3 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 B^{-1}A^{-1} &= \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & -2/3 \\ 0 & 1/3 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0 & 1/3 \end{bmatrix}
 \end{aligned}$$

Problem 2:

$$[A | I]$$

$$= \left[\begin{array}{cccc|ccc} 0 & 2 & 4 & 0 & 1 & 0 & 0 \\ 1 & 1 & 6 & -1 & 0 & 1 & 0 \\ 3 & 4 & 2 & 1 & 0 & 0 & 1 \end{array} \right]$$

$3 \times 3 \quad 3 \times 4$

$$= \left[\begin{array}{cccc|ccc} 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 & 1 & 0 & 0 \\ 3 & 4 & 2 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{cccc|ccc} 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2 & 4 & 0 & -9 & 1 \end{array} \right]$$

$$= \left[\begin{array}{cccc|ccc} 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 4 & -1/2 & -3 & 1 \end{array} \right]$$

$$= \left[\begin{array}{cccc|ccc} 1 & 0 & -2 & 0 & 5/8 & 1 & 1/4 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 4 & -1/8 & 0 & 1/4 \end{array} \right]$$

A E

$$EA = R$$

$$\left[\begin{array}{ccc|ccc} -5/8 & 1 & 1/4 & 0 & 2 & +2 & 0 \\ 1/2 & 0 & 0 & 1 & 1 & 0 & -1 \\ -1/8 & 0 & 1/4 & 3 & 4 & 2 & 1 \end{array} \right]$$

$$= \left[\begin{array}{cccc} 1 & 0 & -2 & -1 \\ 0 & 1 & 2 & 0 \end{array} \right]$$

$$= \left[\begin{array}{cccc|ccc} 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 4 & -1/2 & -3 & 1 \end{array} \right]$$

$$= \left[\begin{array}{cccc|ccc} 1 & 0 & -2 & -1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1/8 & -3/4 & 1/4 \end{array} \right]$$

$$= \left[\begin{array}{cccc|ccc} 1 & 0 & -2 & 0 & -5/8 & 1/4 & 1/4 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1/8 & -3/4 & 1/4 \end{array} \right]$$